

Caribbean Coral Reef Institute (CCRI) Water Quality Monitoring Lab University of Puerto Rico – Mayagüez Campus Department of Marine Sciences			
Document Title	Document Number	Version	
SOP Total o-Phosphate	WQP 012	1.4	

PREPARED BY			
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VERSION HISTORY			
Version	Date	Description	Approved By
1.0	5/10/23	Initial Release for quantifying total ortho-phosphate	TAC
1.1	9/8/23	Increased sample volume and changed to acid addition in the lab according to QAPP.	TAC
1.2	11/30/23	SOP instructions revised according to chemical reagent changes in pH. Sample preservation changes in acid and base volumes addition. Addition of spike preparation instructions and CRM preparation instructions updated to improve detection.	TAC
1.3	6/10/24	Modification of CRM preparation instructions on Section 13 to include combined ERA Waters CRM. Addition of Spike preparation instructions to include a total volume of 10mL sample/blank. The instrument preparation section was moved before running the instrument start-up. Addition of instructions to retrieve report as ASCII file. Revision of the acidification and neutralization instructions.	TAC
1.4	9/4/24	Sample neutralization volume changed to 500 uL of NaOH for the 100 mL sample for more consistent results. The option to neutralize the 15 mL sample was eliminated.	TAC

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1. Title
- Standard Operating Procedure for the Determination of Total Phosphates in Seawater
2. Applicability
- This SOP applies to the determination of Total Phosphates at shallow coral reef sites around Puerto Rico.
3. Purpose
- 3.1. Provide laboratory-specific instructions for the CCRI Water Quality Monitoring Lab personnel to measure coral reef seawater Total Phosphates at shallow coral reef sites around Puerto Rico.

3.2. Provide water quality data to be uploaded to the EPA WQX database for use by state and federal agencies and any other parties interested in coral reef Total Phosphate data following the Quality Assurance Project Plan.
4. Method
- 4.1. EPA Method 365.1 (Environmental Protection Agency)
5. Health and Safety
- 5.1. Follow the safety indications for equipment and personnel suggested by the manufacturer.

5.2. Read each reagent SDS prior to use for safe handling and disposal.

5.3. Prepare reagents to be used on this SOP inside a hood.

5.4. Wear disposable latex gloves, lab coat, and safety glasses while performing this procedure.

5.5. Be aware of waste container liquid level while analyzing samples to avoid spills.
6. General Precautions
- 6.1. Always maintain two points of contact when deploying the Van Dorn bottle on the boat to avoid losing the bottle overboard.

6.2. Ensure the boat engines are idle and the captain approves of the Van Dorn bottle being deployed overboard before deploying the bottle to ensure that the line does not become entangled with the boat propeller.

6.3. Strictly follow SOPs and manufacturer instructions.

6.4. Keep all instrument covers in place during analysis.

6.5. Ensure the supply of reagents and wash is adequate to complete the run.

6.6. Never attempt to turn off or change pump speed during a run, unless there is an emergency.

6.7. Never leave the pump platen engaged without pumping for long periods of time.

6.8. Ensure the waste tube is never submerged in the waste.
7. Training
- 7.1. Personnel shall be trained in the proper sample collection, handling, and analysis procedure for Total Phosphates.

7.2. Personnel shall receive training on how to complete the field sampling log for this SOP.
8. Required Materials
- 8.1. Field Sampling Log Form (Appendix F)

8.2. Consumables

8.2.1. Kimwipes

8.2.2. Latex Gloves

8.2.3. Beaker (1000 mL) for solution preparation

8.2.4. Graduated cylinder (1000 mL, 200 mL)

8.2.5. Plastic/Glass bottles for reagent and wash storage (1000 mL)

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- 8.2.6. Plastic Bottles for reagent and stock standard storage (100 mL)
 8.2.7. Amber plastic bottles for reagent storage (1000 mL)
 8.2.8. 125 mL polypropylene plastic bottle
 8.2.9. Plastic Syringe for Filtering Sample
 8.2.10. PES Filters (0.45 µM) 1 per seawater sample
 8.3. AA500 AutoAnalyzer
 8.4. Required Chemicals:

Compound	CAS No.	Safety classification	
Acetone	67-64-1	GHS02, GHS07	Danger – Flammable, Irritant
Ammonium heptamolybdate	12054-85-2	GHS07	Irritant
Antimony potassium tartrate	331753-56-1	GHS07, GHS09	Irritant, Warning/Danger - Environmentally Damaging
Ascorbic acid	50-81-7	--	
Potassium dihydrogen phosphate	7778-77-0	--	
Sodium chloride	7647-14-5	--	
Sodium hypochlorite, 10% solution	7681-52-9	GHS05, GHS09	Warning/Danger – Corrosive, Warning/Danger - Environmentally Damaging
Sodium hydroxide	1310-73-2	GHS05	Warning/Danger – Corrosive
Sulfuric acid, conc.	7664-93-9	GHS05	Warning/Danger – Corrosive

- 8.5. Reagents, Standards, and Preparation Instructions
 8.5.1. Synthetic seawater

Reagent	Amount	Instructions
Sodium chloride	35 g	1. Weigh sodium chloride. 2. Add the measured amount into a beaker with a stir rod. 3. Add approximately 500-600 mL of distilled water to the beaker. 4. Stir on a plate until all the salt has dissolved. 5. Pour beaker contents into a 1L graduated cylinder. 6. Bring to volume with distilled water.
Distilled water	1L	
General details		
Storage	Plastic container at room temperature.	
Stability	One week.	
Purpose	Stock standard solution preparation.	

- 8.5.2. Hydrochloric Acid, 1N

Reagent	Amount	Instructions	
Hydrochloric acid 36.5-38%	85 mL	1. Add approximately 700 mL of distilled water into a graduated cylinder. 2. Measure HCl in a 100 mL graduated cylinder and slowly add it to step #1. 3. Let it cool to room temperature. 4. Bring to volume with distilled water. 5. Transfer to a storage container and mix.	
DistilledDI water	1000 mL		
General Details			
Storage	Glass container at room temperature.		
Stability	Indefinite. Replace if turbid.		

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Purpose	For weekly cleanup.
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8.5.3. Sodium hydroxide (1 mol/L)		
Reagent	Amount	Instructions
Sodium hydroxide	40 g	1. Weigh the NaOH. 2. Add the weighted material to a beaker with approximately 500 mL of distilled water. 3. Place the beaker on a stir plate and mix the contents. 4. Transfer the solution to a graduated cylinder and bring to volume with distilled water.
Distilled water	1000 mL	
General details		
Storage	Plastic container at room temperature.	
Stability	Indefinite. Replace if turbid.	
Purpose	For weekly cleanup.	

8.5.4. Special Wash Solution		
Reagent	Amount	Instructions
Clorox Bleach (Sodium hypochlorite 10% solution)	10 mL	1. Dilute 10 mL of Clorox bleach into 50 mL of distilled water.
Distilled water	50 mL	
General details		
Storage	Plastic container at room temperature	
Stability	One Day.	
Purpose	For weekly cleanup.	

8.5.5. Stock Antimony Potassium Tartrate		
Reagent	Amount	Instructions
Antimony potassium tartrate	1.15 g	1. Weigh the Antimony potassium tartrate. 2. Add the weighted material to a beaker with approximately 50 mL of distilled water. 3. Place the beaker on a stir plate and mix the contents. 4. Transfer the solution to a graduated cylinder and bring to volume with distilled water.
Distilled water	50 mL	
General details		
Storage	Plastic container at room temperature.	
Stability	One month.	
Purpose	For Ammonium molybdate reagent preparation.	

8.5.6. Ammonium molybdate		
Reagent	Amount	Instructions
Sulfuric acid, conc.	16 mL	1. Add approximately 500mL of distilled water to a beaker. 2. Measure the sulfuric acid and carefully add it to the beaker. 3. Weigh the ammonium heptamolybdate and add it to the beaker. 4. Place the beaker on a stir plate and mix the contents. 5. Add the stock antimony potassium tartrate to the beaker.
Ammonium heptamolybdate	1.5 g	

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Stock antimony potassium tartrate	5.5 mL	6. Pour contents into a graduated cylinder and bring to volume with distilled water.
Distilled water	500 mL	
General details		
Storage	Amber plastic container at 4°C.	
Stability	One week.	
Purpose	For PO ₄ analysis.	

8.5.7. Ascorbic Acid

Reagent	Amount	Instructions
Ascorbic acid	1 g	1. Weigh the ascorbic acid. 2. Add the weighted material to a beaker with approximately 600 mL of distilled water. 3. Place the beaker on a stir plate and mix the contents. 4. Add the acetone to the beaker. 5. Pour contents into a graduated cylinder and bring to volume with distilled water.
Acetone	5 mL	
Distilled water	500 mL	
General details		
Storage	Amber plastic container at 4°C.	
Stability	One week.	
Purpose	For PO ₄ Analysis.	

8.5.8. Stock Standard A (1000 mg/L) (PO₄-P)

Reagent	Amount	Instructions
Potassium dihydrogen phosphate	0.4393 g	1. Weigh the potassium dihydrogen phosphate. 2. Add the weighted material to a beaker with approximately 60 mL of distilled water. 3. Place the beaker on a stir plate and mix the contents. 4. Pour contents into a graduated cylinder and bring to volume with synthetic seawater.
Synthetic seawater	100 mL	
General details		
Storage	Plastic container at 4°C.	
Stability	One year.	
Purpose	For working standards preparation.	

8.5.9. Stock Standard C (100 μmol/L)

Reagent	Amount	Instructions
Stock standard A	0.31 mL	1. Pipette 0.31 mL of stock standard A. 2. Pour volume into a 100 mL volumetric flask. 3. Bring to volume (100 mL) with synthetic seawater.
Synthetic seawater	100 mL	
General details		
Storage	Plastic container at 4°C.	
Stability	Three months.	
Purpose	For working standards (Low Range) preparation. Necessary if PO4-only analysis will be performed. Otherwise, Stock Standard A is	

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	mixed with the NOx Stock Standard A to create a combined standard for simultaneous NOx and o-PO4 analyses.
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8.5.10. Working Standards

Prepare fresh working standards before each analysis run.

- 8.5.10.1. Combined Stock standard [5 mg/L] - **This is to be prepared if running both NOx and PO4 analyses simultaneously (both instrument channels are used).**
- 8.5.10.1.1. Pipette 500 μL (0.5 mL) of Nitrate Stock Standard A (1000 mg/L) into a 100 mL volumetric cylinder.
- 8.5.10.1.2. Pipette 500 μL (0.5 mL) of Phosphate Stock Standard A (1000 mg/L) into the same 100 mL volumetric cylinder.
- 8.5.10.1.3. Bring the volumetric flasks to volume with **synthetic seawater** and mix thoroughly.
- 8.5.10.1.4. Prepare calibration standards in individual volumetric flasks using the following table:

Combined Stock (mL)	Combined Stock (μL)	Combined Calibrants [mg/L]
4.0	4,000	0.200
2.0	2,000	0.100
1.0	1,000	0.050
0.5	500	0.025
0.2	200	0.010
0.1	100	0.005

8.5.10.2. PO4 Only Channel Run

- 8.5.10.2.1. Pipette standard C into a 100 mL volumetric flask, following the table below.
- 8.5.10.2.2. Dilute to volume with **synthetic seawater** and mix thoroughly.

mL of Stock Standard C (Low Range)	μL of Stock Standard C (Low Range)	Calibrants [mg/L]
2.0	2,000	0.062
1.5	1,500	0.046
1.0	1,000	0.031
0.5	500	0.015
0.2	200	0.006

9. Procedure

9.1. Sample Collection

- 9.1.1. Prepare a Van Dorn bottle to be deployed.
- 9.1.2. Lower the Van Dorn bottle in a slow and steady mode until it gets close to the bottom with caution not to touch the bottom and stir sediment up.
- 9.1.3. Drop the bottle messenger to close the Van Dorn bottle at the desired depth.
- 9.1.4. Pull the Van Dorn bottle back up into the boat.
- 9.1.5. Connect the tubing to the Van Dorn bottle spigot to transfer water.
- 9.1.6. Push open the Van Dorn bottle’s spigot valve to allow water flow.
- 9.1.7. Dispense a few mL of the seawater into a clean 125 mL polypropylene plastic bottle to rinse the bottle 3 times.

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- 9.1.8. Fill the 125 mL polypropylene plastic bottle with seawater.
- 9.1.9. Place the sample bottle inside the cooler until analysis in the lab.
 - 9.1.9.1. Samples can be preserved for 28 days at 4°C prior to analysis.
 - 9.1.9.2. **Poison the sample** upon arrival in the lab.
 - 9.1.9.2.1. Carefully add 400 µL of 5.4 N sulfuric acid (H₂SO₄) to a 100 mL seawater sample bottle.
 - 9.1.9.2.2. Close the Falcon tube and invert to mix.
 - 9.1.9.3. Analyze samples within 28 days.
 - 9.1.9.4. **Prior to analysis, the sample should be brought to neutral pH** by adding 500 µL of 5.4 N NaOH solution.

9.2 Sample/Instrument Pre-Run Preparation: Analysis Setup

- 9.2.1 Prepare Sampler tubes according to the Analysis Protocol with calibration standards and previously filtered samples.
 - 9.2.1.1 Pour the highest calibration standard into a 50 mL tube and position it into the calibrant rack to the left-hand side (Numbers 921 or 922).
 - 9.2.1.2 Pour the remaining calibration standards into the 10 mL small sampler cups and position them into the calibrant rack to the left-hand side (Numbers 901-920).
 - 9.2.1.3 **Filter samples to be analyzed** directly into the 10 mL sampler cups and position them into the sample racks on the right-hand side (Numbers 1-120).
 - 9.2.1.3.1 Note: ideally one clean filter will be used for each sample, but filters may be re-used with 5 mL rinses of sample water before saving the filtrate for analysis.
- 9.2.2 Create a New Run or Copy Run
 - 9.2.2.1 Click “Set Up”
 - 9.2.2.2 Click “Analysis/Run”
 - 9.2.2.3 Double-click on the “NOx_PO4” folder from the left side of the pop-up window.
 - 9.2.2.4 Select the NOx_PO4.anl run from the list in the center of the window.
 - 9.2.2.5 Click “New Run” or “Copy Run” to create the run based on the previously selected run template.
 - 9.2.2.5.1 Insert Run Name: “Date_Quarter#_NPRun#” ” on the provided space at the upper right-hand corner of the Main tab. (E.g. 20231025_Q2_NP1)
 - 9.2.2.5.2 Check Channel tab
 - 9.2.2.5.2.1 Calibrants concentrations and units (mg/L).
 - 9.2.2.5.2.2 Spike Recovery Standard Concentration [5 mg/L]
 - 9.2.2.5.3 Modify the Tray Protocol tab to match your run according to the following criteria:
 - 9.2.2.5.3.1 1 Primer (P) and 1 Drift (D) with the highest calibrant.
 - 9.2.2.5.3.2 Calibrants + Blank (C) in decreasing order.
 - 9.2.2.5.3.3 Baseline (B) whose cup number is 0.
 - 9.2.2.5.3.4 3 Carryovers; 1 H1 using the highest calibrant and 2 L2 using the lowest calibrant.
 - 9.2.2.5.3.5 1 Recovery - Spiked Blank (K) - See Section 13 for Spike preparation instructions.
 - 9.2.2.5.3.6 1 CRM (NO₃) as QC1 - See Section 13 for CRM preparation instructions.
 - 9.2.2.5.3.7 10 Samples
 - 9.2.2.5.3.8 1 Spiked Sample (K) - See Section 13 for Spike preparation instructions.
 - 9.2.2.5.3.9 1 CRM as QC1 (NO₃)

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- 9.2.2.5.3.10 Baseline (B) whose cup number is 0.
- 9.2.2.5.3.11 Repeat 10 samples, 1 QC's, and 1 Baseline until completing the sample analysis for the day.
- 9.2.2.5.3.12 Ensure that the tray protocol ends with 1 Drift (D) using the highest calibrant, and 1 End (E), which is the cup number 0.
- 9.2.2.5.4 Other ways to create a New run based on previously existent Tray Protocol:
 - 9.2.2.5.4.1 Import a previously created tray protocol from Excel. Make sure the tray protocol follows the correct format to be imported.
 - 9.2.2.5.4.2 Excel file must be closed prior to importing.
- 9.2.2.5.5 Click "OK" to save the new run.

9.3 Autoanalyzer Instrument Start-Up

- 9.3.1 Ensure you have an adequate supply of reagents to complete the run.
- 9.3.2 Turn ON the instrument in the following order.
 - 9.3.2.1 Computer
 - 9.3.2.2 Sampler
 - 9.3.2.3 Power Supply
- 9.3.3 Open the AACE Software.
- 9.3.4 Login - No password
 - 9.3.4.1 Click "OK".
- 9.3.5 Confirm the analyzer and software are communicating by observing a permanent blue light in the photometer module, meaning it is ready for operation.
 - 9.3.5.1 Blinking blue light means the system is operating (charting & running).
 - 9.3.5.2 Red - STOP! Leak detected.
- 9.3.6 Place all lines inside the distilled water bottle.
 - 9.3.6.1 Ascorbic acid line - blue line labeled "PO4 Ascorbic"
 - 9.3.6.2 Ammonium molybdate - red line labeled "PO4 Molybdate"
 - 9.3.6.3 Water + SDS - clear line labeled "PO4 Water + SDS"
 - 9.3.6.4 Wash line - clear thick line labeled "Channel 3 Wash"
 - 9.3.6.5 Rigid clear line labeled "PO4 Water"
- 9.3.7 Place the sampler wash line into the distilled water bottle.
 - 9.3.7.1 This line is the only one to the right side of the AS2 sampler (close to the computer station).
- 9.3.8 Click "Charting" at the system window toolbar.
 - 9.3.8.1 Or click on "Run" in the main menu and select "Chart"
 - 9.3.8.2 Click "OK" if the NOx_PO4 analysis is displayed or "Change" to select NOx_PO4 analysis.
- 9.3.9 Click on the "Reagent Valve Set 1" icon on the system's main window.
 - 9.3.9.1 Select "Ch1_Ch3.rva".
 - 9.3.9.2 Click "OK".
- 9.3.10 Click on the "Pump" icon on the system's main window.
 - 9.3.10.1 Select "Normal" to pump at normal speed.
 - 9.3.10.2 Select "Engage Platen".
 - 9.3.10.2.1 Observe the platen moving down slowly. When in place, the LED light on the control switch will turn green.
 - 9.3.10.3 Press "Close" to close the window.
 - 9.3.10.3.1 Pump will continue running at the selected speed.
- 9.3.11 Set Light Power.
 - 9.3.11.1 Perform this adjustment every time an analysis is run for the first time when pump tubes get changed, and occasionally to compensate for lamp aging.
 - 9.3.11.2 Right-click on the charting window.

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- 9.3.11.3 Select “Set Light Power” from the pop-up menu.
- 9.3.12 Ensure the Gain is set to 10.
 - 9.3.12.1 Gain value can be seen at the bottom of the charting window.
 - 9.3.12.2 If not set to 10, change the value by clicking on “Channel 2” on the system’s main window and manually inserting the right value in the provided space for the Gain.
- 9.3.13 Perform Water Baseline Check.
 - 9.3.13.1 Pump DI water through the system for 10-15 minutes at normal speed.
 - 9.3.13.1.1 Monitor the flowchart and wait for the system to stabilize.
 - 9.3.13.1.2 Visually inspect the bubble pattern through the whole system for the same size, shape, and distance. See AA500 Complete page 6.6 for bubble visual reference.
 - 9.3.13.2 **Set the base** once the system is stable.
 - 9.3.13.2.1 Right-click on the chart.
 - 9.3.13.2.2 Select “Set Base/Offset”.
 - 9.3.13.2.3 Refer to “Wash Solution Baseline” in the “ How To” Section if baseline does not stabilize.
 - 9.3.13.3 Note the water baseline value (%) in the chart’s upper right corner.
 - 9.3.13.3.1 Expected value: 5.0%
- 9.3.14 Perform Reagent Check.
 - 9.3.14.1 Remove PO4 Ascorbic and PO4 Molybdate reagent lines from the distilled water bottle.
 - 9.3.14.2 Pat dry the reagent line tubes with a Kimwipe.
 - 9.3.14.3 Insert reagent lines into their corresponding reagent solution containers.
 - 9.3.14.3.1 “PO4 Ascorbic” into the Ascorbic Acid bottle.
 - 9.3.14.3.2 “PO4 Molybdate” into the Ammonium Molybdate bottle.
 - 9.3.14.3.3 Refer to the system diagrams in case of doubt.
 - 9.3.14.4 Leave the remaining lines in the distilled water bottle.
 - 9.3.14.5 Continue pumping solutions through reagent lines at normal speed for 10-15 minutes.
 - 9.3.14.5.1 Visually inspect the bubble pattern and allow it to stabilize.
 - 9.3.14.6 Annotate the reagent baseline value (%) on the chart’s window’s upper right corner.
 - 9.3.14.7 Calculate the reagent absorbance.
 - 9.3.14.7.1 Insert the reagent baseline value (%) and the gain (10) into the corresponding space in the “Absorbance_Calcs.” worksheet located in the following path: Desktop>AA500 Docs&Manuals>DailyUse
 - 9.3.14.7.2 If the resulting absorbance is >20%, the reagents and/or wash pot may be contaminated.
 - 9.3.14.8 Wait several minutes until a peak comes out and ensure the baseline is stable.
 - 9.3.14.9 **Reset the Base.**
 - 9.3.14.9.1 Right-click on the chart screen.
 - 9.3.14.9.2 Select “Set Base/Offset”.
 - 9.3.14.10 Wait several minutes to ensure the baseline is stable.
- 9.3.15 Perform Sensitivity (Gain) Check.
 - 9.3.15.1 Sample the highest calibrant
 - 9.3.15.1.1 Double-click on the “Sampler” icon.
 - 9.3.15.1.2 Enter the cup number containing the highest calibrant.
 - 9.3.15.1.3 Click on “Sample”.
 - 9.3.15.1.4 Pump for 2 minutes.
 - 9.3.15.1.5 After 2 minutes, select “Wash”.
 - 9.3.15.1.6 Close the window by clicking “Cancel”.

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- 9.3.15.2 **Observe the Primer peak (1st peak) until it reaches a plateau.**
 - 9.3.15.2.1 This may take a few minutes.
 - 9.3.15.2.2 Annotate the peak value %.
- 9.3.15.3 Right-click on the chart screen and select **“Set Gain”** while the peak is still at the plateau.
- 9.3.15.4 Calculate the Sensitivity
 - 9.3.15.4.1 Complete the Sensitivity table in the “Absorbance_Calcs.” worksheet located in the following path: Desktop>AA500 Docs&Manuals>DailyUse.
 - 9.3.15.4.2 The peak % used is the one observed after setting the gain (~90%)
- 9.3.15.5 **Wait until the peak fully passes and reset the base.**
 - 9.3.15.5.1 Right-click on the chart screen and select **“Set Base/Offset”**.
- 9.3.15.6 Verify the baseline and bubble pattern stability.
- 9.3.16 Proceed to analyze samples.

9.4 Start the Run

- 9.4.1 **Verify that all sample IDs refer to the correct sample rack position in the Tray Protocol before proceeding.**
- 9.4.2 Click on “Run” in the main menu.
- 9.4.3 Click on “Start”.
- 9.4.4 Double-click on the “Data” folder under the “Analysis” area.
- 9.4.5 Double-click on “NOx-PO4”.
- 9.4.6 Select the previously created .run file from the list in the center of the window.
- 9.4.7 Click “OK”.
- 9.4.8 Insert the analyst's name and comments in the provided spaces on the pop-up window.
- 9.4.9 Click “OK” to start the run.
 - 9.4.9.1 The sample protocol will start running and will stop once the last command on the Tray Protocol has been run.
 - 9.4.9.2 Wait for a pop-up window with a run completion message on the top left corner of the computer window.
- 9.4.10 Continue with Daily Shutdown instructions in Section 9.5.

9.5 Daily Shutdown

- Perform this Daily shutdown if the instrument will be used for a few consecutive days otherwise, complete a Weekly Cleanup (Refer to Section 11 for instructions.)
- 9.5.1 Restart Charting.
 - 9.5.1.1 Once the system ends a run, it closes the charting windows and disables some software functions.
 - 9.5.2 Leave the sampler wash line in the distilled water bottle.
 - 9.5.3 Remove the reagent lines from the reagent bottles.
 - 9.5.4 Pat dry the tubes with a Kimwipe.
 - 9.5.5 Place the reagent lines into the distilled water bottle.
 - 9.5.6 Pump for a minimum of 20 minutes.
 - 9.5.7 After 20 minutes, remove the lines from the distilled water bottle.
 - 9.5.7.1 Drain the system by continuing pumping. Observe the lines in the chemistry module until most bubbles have exited the system.
 - 9.5.8 Pat dry the tubes and store them inside the provided flasks.
 - 9.5.9 Turn off the pump and release the platen.
 - 9.5.9.1 Click on the “Pump” icon on the main window.
 - 9.5.9.2 Switch the pump off.
 - 9.5.9.3 Click on “Release platen”.

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- 9.5.10 Stop Charting.
 - 9.5.10.1 Click the “Stop” button in the System window toolbar.
 - 9.5.10.1.1 Or click the “Close” button in the charting window.
- 9.5.11 Turn OFF the instrument in the following order.
 - 9.5.11.1 Power Supply
 - 9.5.11.2 Sampler
 - 9.5.11.3 Computer
- 9.5.12 Discard samples and calibrants from the Sampler cups in the provided waste container.
- 9.5.13 Wash sample cups.
 - 9.5.13.1 Rinse cups with tap water 3 times.
 - 9.5.13.2 Submerge the cups in a 10% HCl bath and let soak overnight.
 - 9.5.13.3 After use, cover and save the HCl bath inside a fume hood to be reused several times.
 - 9.5.13.3.1 Discard into the corresponding unwanted material container if turbid.
 - 9.5.13.4 Triple-rinse the sample cups with DI water.
 - 9.5.13.5 Let the cups dry standing in a dish rack.

9.6 Data Retrieval

- 9.6.1 Reports
 - 9.6.1.1 Click on “Retrieve” on the main menu.
 - 9.6.1.2 Select “Report” from the dropdown.
 - 9.6.1.3 Double-click on the desired analysis (menu on the left window).
 - 9.6.1.4 Double-click on the desired .run File.
 - 9.6.1.5 Select the desired option from the buttons at the bottom of the report window.
 - 9.6.1.5.1 You may save a PDF version of the report at this point.
- 9.6.2 ASCII files
 - 9.6.2.1 Click on “Retrieve” on the main menu.
 - 9.6.2.2 Select “View Chart” from the dropdown.
 - 9.6.2.3 Double-click on the desired analysis (menu on the left window).
 - 9.6.2.4 Double-click on the desired .run File.
 - 9.6.2.5 On the icons on top of the chart window, click on the “Export to ASCII file” icon (3rd left to right)
 - 9.6.2.6 Format Tab
 - 9.6.2.6.1 Format should always be set to “Text Format (ASCII)”
 - 9.6.2.6.2 Confirm the Directory to be used
 - 9.6.2.7 Export Data Tab
 - 9.6.2.7.1 Check all desired information to be included in the file
 - 9.6.2.7.2 Go to the “Preview Tab” to confirm the selection
 - 9.6.2.8 Click on “Export” to save the file.

10. Documentation

- 10.1. Field data will be recorded in the Sampling Log (Appendix F).
- 10.2. Field forms and Maintenance sheets shall be kept in the designated file cabinet at the CCRI office.
- 10.3. Daily/Weekly/Monthly/Yearly maintenance shall be recorded in maintenance sheets.
- 10.4. Data contained in hard copy forms shall be digitized into the corresponding electronic spreadsheets as soon as possible after the task is completed.
- 10.5. All spreadsheets should be uploaded to the internal CCRI Water Quality Database.
- 10.6. All data will be electronically archived and backed up online.

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11. Cleaning and Maintenance

- 11.1. The autoanalyzer exterior may be cleaned using an alcohol-damped lint-free cloth.
- 11.2. Refer to “AA500 Complete User Manual” for specific instrument maintenance instructions.
- 11.3. Recommended pump tubes use is for a maximum of 200 hours.
- 11.4. Autoanalyzer Daily Cleanup
 - 11.4.1. Follow the instructions provided under Section 9.5 Daily Shutdown.
 - 11.4.2. Document your cleanup method in the provided lab notebook.
- 11.5. Autoanalyzer Weekly Cleanup
 - 11.5.1. Select “All Wash” Mode by pressing the reagent valve set 1 icon
 - 11.5.2. Pump the following through all reagent lines for at least 2 minutes each.
 - 11.5.2.1. NaOH (Sodium hydroxide)
 - 11.5.2.2. Distilled water
 - 11.5.2.3. HCl 1N (Hydrochloric acid)
 - 11.5.2.4. Distilled water
 - 11.5.2.5. Special Wash Solution 10% (Sodium hypochlorite)
 - 11.5.2.6. Distilled water
 - 11.5.3. Remove the lines from the distilled water bottle and store them inside their respective channel storage flask.
 - 11.5.4. Leave the pump running for a few minutes to drain the system.
 - 11.5.4.1. Observe until no liquid is seen inside the system and/or the waste line.
 - 11.5.5. Once the system has drained, stop Charting.
 - 11.5.5.1. Click the “Stop” button in the System window toolbar.
 - 11.5.5.1.1. Or click the “Close” button in the charting window.
 - 11.5.6. Print and fill out the Weekly Maintenance Sheet included in the user manual.
 - 11.5.6.1. Save the sheet in the instrument’s maintenance binder.
- 11.6. Autoanalyzer Monthly Cleanup
 - 11.6.1. Disconnect the cadmium column from the flow.
 - 11.6.2. Wash the manifold and sample line with the special wash solution (diluted hypochlorite solution) for 10 mins.
 - 11.6.3. Follow the previous wash with distilled flushing all reagent lines for 10 minutes.
 - 11.6.4. Print and fill out the Monthly Maintenance Sheet included in the user manual.
 - 11.6.4.1. Save the sheet in the instrument’s maintenance binder.
- 11.7. For Monthly and Yearly Maintenance follow the instructions provided in the AA500 User Manual and complete the corresponding documentation.
 - 11.7.1. AA500 Operation Manual - Section 5
 - 11.7.2. AA500 Complete User Manual - Section 4

12. Troubleshooting

- 12.1. Refer to AA500 Complete User Manual (Section 8) For Troubleshooting instructions.
- 12.2. Refer to AA500 Complete User Manual (Section 6.10) for sensitivity and noise troubleshooting instructions.

13. QA/QC

- 13.1. As per QAPP Table 4, seawater Total Phosphates should be measured in mg/L with an accuracy of 0.004 mg/L (±20%) and a precision of <20%.
- 13.2. Any seemingly erroneous measurements will be flagged for further review in accordance with the QAPP.
- 13.3. CRM Preparation
 - 13.3.1. Make CRM dilution fresh before each analysis.
 - 13.3.2. Phosphate - Pipette 7,500 µL (7.5 mL) of Phosphate CRM [0.400 mg/L] solution and bring to volume with synthetic seawater in a 100 mL volumetric flask. This dilution will have a concentration of 0.030 mg/L.
 - 13.3.3. Combined ERA CRM

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- 13.3.3.1. Initial Dilution as instructed by the manufacturer- Pipette 2500 µL (2.5 mL) of previously concentrated ERA CRM vial and bring to volume with **synthetic seawater** in a 500 mL volumetric flask. This dilution will have a concentration of 23 mg/L.
- 13.3.3.2. Pipette 650 µL (0.65 mL) of previously diluted ERA CRM [23 mg/L] and bring to volume with **synthetic seawater** in a 100 mL volumetric flask. This dilution will have a concentration of 0.03 mg/L.
- 13.4. Spike Preparation (total vol. = 5 mL)
 - 13.4.1. Pipette 4,850 µL of Synthetic Sea Water/Sample into a sample cup
 - 13.4.2. Add 150 µL (0.15 mL) of Combo Stock Solution [5 mg/L] into the sample cup
 - 13.4.3. Cap the tube and gently invert it to mix the contents.
 - 13.4.4. This will add 0.15 mg/L O-PO4 to the sample.
- 13.5. Spike Preparation (total vol. = 10 mL)
 - 13.5.1. Pipette 9,970 µL of Synthetic Sea Water/Sample into a sample cup
 - 13.5.2. Add 300 µL (0.30 mL) of Combo Stock Solution [5 mg/L] into the sample cup
 - 13.5.3. Cap the tube and gently invert it to mix the contents.
 - 13.5.4. This will add 0.15 mg/L O-PO4 to the sample.

14. References

- 14.1. U.S. ENVIRONMENTAL PROTECTION AGENCY. 1993. Methods for Determination of Inorganic Substances in Environmental Samples; EPA-600/R/93/100—Draft. Environmental Monitoring Systems Lab., Cincinnati, Ohio.
- 14.2. AA500 Complete User Manual.
- 14.3. AA500 Operational Manual.
- 14.4. AACE Software Manual.